

eacher System Gap Analysis & Implementation Plan

Executive Summary

The teacher system has substantial backend infrastructure (~75% complete) but zero teacher-facing frontend (0% complete). Admin can manage teachers/teaching events, but teachers cannot self-serve.

Current State Assessment

Backend Completeness

Component: Teacher Model

Status: 95%

Notes: Rich schema, payment settings, subscription integration

Component: TeachingEvent Model

Status: 95%

Notes: Complex scheduling, registration config, affiliate fields disabled

Component: TeacherSubscription Model

Status: 100%

Notes: Full subscription lifecycle

Component: teacher.controller.ts

Status: 70%

Notes: Many stubs (media upload, availability, social links)

Component: teacher.service.ts

Status: 20%

Notes: Only 1 function! Logic in controllers

Component: Payment controllers

Status: 75%

Notes: Stripe Connect started, subscription unclear

Component: Payout controllers

Status: 80%

Notes: Dashboard, history, requests implemented

Component: Admin controllers

Status: 60%

Notes: CRUD defined, implementation unclear

Frontend Completeness

Component	Status	Notes
Admin Teaching Events	80%	Full CRUD, bulk actions, 5-tab editor
Admin Teacher Management	40%	List/approve exists, no detailed management
Teacher Dashboard	0%	Does not exist
Teacher Profile Page	0%	Does not exist
Teacher Event Management	0%	Does not exist
Teacher Bookings/Students	0%	Does not exist
Teacher Payments/Payouts	0%	Does not exist
Teacher Analytics	0%	Does not exist

Critical Bugs to Fix First

P0 - Blocking Issues

- Refund Transaction Bug - teaching_event.cancellation.controller.ts:98

- Refunds happen outside main transaction
 - If refund fails, event marked cancelled but money not returned
 - Fix: Wrap batch refund in same transaction
2. Affiliate System Disabled - TeachingEvent.ts:133-669
- Fields commented out: isAffiliateEvent, claimStatus, claimedBy
 - Controllers use (event as any) casting
 - Decision needed: Enable or remove
3. Email Templates Not Wired - teaching_event.registration.controller.ts
- queueEmail() called but templates don't exist
 - Registrations, cancellations, approvals don't notify users
4. Function Import Mismatches - teacher.routes.ts
- Routes import uploadTeacherMedia but controller has uploadTeacherImage
 - Routes import updateTeacherAvailabilityHours but controller has updateTeacherAvailability
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Missing Backend Features

Teacher Profile (teacher.controller.ts)

Stubbed/Incomplete:

- updateTeacherProfile (lines 650-729) - Many fields commented out:
 - subjects, expertise, qualifications, certifications
 - website, teachingDescription, teachingAvailability
- uploadTeacherImage (line 745) - No implementation
- updateTeacherAvailability (line 785) - Save logic commented
- updateTeacherSocialMedia (line 833) - Save logic commented

Missing:

- Demo video upload
- Portfolio/gallery management
- Certificate upload for qualifications
- Bio rich text support

Teacher Service Layer (teacher.service.ts)

Current State: Only has attachOfflineDemoVideo() function

Missing Services:

- updateTeacherProfile() - Profile update business logic
- verifyTeacherQualifications() - Document verification workflow
- calculateTeacherStats() - Aggregate stats calculation
- getTeacherEarnings() - Revenue aggregation
- archiveTeacher() - Soft delete with cascading

Payment/Payout Services

Partially Implemented:

- teacher.payment.service.ts - Status unclear
- teacher.payout.service.ts - Partial (80 lines visible)
- teacher.stripe-connect.service.ts - Status unclear
- teacher.subscription.service.ts - Status unclear

Missing:

- Subscription plan upgrade/downgrade
- Proration calculation
- Grace period handling
- Failed payment retry logic

Teaching Event System

Disabled Features:

- Affiliate system (all fields commented)

Missing Features:

- Seat reservation/hold during checkout (overbooking risk)
- Auto-archive expired events (lifecycle jobs)
- Cascading delete for registrations
- File upload virus scanning
- Waitlist management
- Early bird/group pricing
- Recurring events
- Co-teachers support

Missing Frontend - Teacher Dashboard

Required Pages

frontend/src/pages/teacher/

- └─ TeacherDashboardPage.tsx # Overview stats, recent activity
- └─ TeacherProfilePage.tsx # Profile management
- └─ TeacherEventsPage.tsx # Teaching events list
- └─ TeacherCreateEventPage.tsx # Event creation wizard
- └─ TeacherEditEventPage.tsx # Event editing
- └─ TeacherStudentsPage.tsx # Enrolled students management
- └─ TeacherBookingsPage.tsx # Booking management
- └─ TeacherRegistrationsPage.tsx # Registration review
- └─ TeacherAnalyticsPage.tsx # Performance analytics
- └─ TeacherPaymentsPage.tsx # Payment settings, Stripe Connect
- └─ TeacherPayoutsPage.tsx # Payout history, requests
- └─ TeacherSubscriptionPage.tsx # Subscription management
- └─ TeacherSettingsPage.tsx # Account settings

Required Components

frontend/src/components/teacher/

- └─ TeacherSidebar.tsx # Navigation sidebar
- └─ TeacherStatsCard.tsx # Dashboard stat cards
- └─ TeacherEventCard.tsx # Event list item
- └─ TeacherStudentList.tsx # Student roster
- └─ TeacherBookingTable.tsx # Bookings table
- └─ TeacherRegistrationCard.tsx # Registration review card
- └─ TeacherPayoutHistory.tsx # Payout history table
- └─ TeacherStripeConnect.tsx # Stripe onboarding flow
- └─ TeacherSubscriptionCard.tsx # Subscription status
- └─ TeacherCalendar.tsx # Event calendar view

Required API Integration

frontend/src/services/api/teacherAPI.ts

- getDashboardStats()
- getProfile() / updateProfile()
- getEvents() / createEvent() / updateEvent() / deleteEvent()
- getBookings() / updateBooking()
- getRegistrations() / reviewRegistration()
- getPayoutDashboard() / requestPayout()
- initiateStripeConnect() / getStripeStatus()
- getSubscription() / paySubscription()

Implementation Phases

Phase 1: Fix Critical Bugs (Week 1)

1. Fix refund transaction isolation bug
2. Enable affiliate system - Uncomment fields in TeachingEvent.ts
3. Wire email templates for notifications
4. Fix route/controller function mismatches
5. Uncomment profile update fields

Phase 2: Complete Backend (Weeks 2-3)

1. Implement teacher.service.ts layer
2. Complete payment/payout services
3. Add approval notification system
4. Implement seat reservation during checkout
5. Add lifecycle jobs for event archival
6. Test affiliate system end-to-end

Phase 3: Teacher Dashboard MVP - Events & Bookings (Weeks 4-6) [PRIORITY]

1. TeacherDashboardPage with overview stats
2. TeacherEventsPage - List, create, edit, delete events
3. TeacherCreateEventPage - Event creation wizard
4. TeacherBookingsPage - View/manage student bookings
5. TeacherStudentsPage - Student roster per event
6. Basic navigation sidebar and layout

Phase 4: Profile & Registration Management (Weeks 7-8)

1. TeacherProfilePage with full edit capability
2. TeacherRegistrationsPage with review flow
3. Registration approval/rejection workflow
4. Student communication features

Phase 5: Payments & Payouts (Weeks 9-10)

1. TeacherPaymentsPage with Stripe Connect onboarding
2. TeacherPayoutsPage with history/requests
3. TeacherSubscriptionPage with plan management
4. Bank account verification UI

Phase 6: Advanced Features (Weeks 11-14)

1. TeacherAnalyticsPage with charts
2. TeacherCalendar component
3. Affiliate dashboard for teachers
4. Student messaging system
5. Attendance tracking
6. Certificate generation

File Modifications Required

Backend Files to Modify

1. backend/src/controllers/teacher.controller.ts
 - Uncomment profile fields (lines 650-729)
 - Implement uploadTeacherImage (line 745)
 - Implement updateTeacherAvailability (line 785)
 - Implement updateTeacherSocialMedia (line 833)
2. backend/src/services/teacher.service.ts
 - Add full service layer methods
3. backend/src/routes/teacher.routes.ts
 - Fix import names to match controller exports
4. backend/src/controllers/teaching_event.cancellation.controller.ts
 - Fix transaction isolation (line 98)
5. backend/src/models/TeachingEvent.ts
 - Enable or remove affiliate fields (lines 133-669)
6. backend/src/services/email.service.ts
 - Add templates for: registration confirmation, approval, rejection, cancellation

Frontend Files to Create

1. All pages in frontend/src/pages/teacher/
2. All components in frontend/src/components/teacher/
3. frontend/src/services/api/teacherAPI.ts

4. frontend/src/types/teacher.ts
 5. frontend/src/hooks/queries/useTeacherQuery.ts
 6. frontend/src/hooks/mutations/useTeacherMutations.ts
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Verification Plan

Backend Testing

Test teacher endpoints

```
npm test -- --grep "teacher"
```

Test teaching event cancellation

```
npm test -- --grep "cancellation"
```

Run full backend tests

```
cd backend && npm test
```

Frontend Testing

Run frontend tests

```
cd frontend && npm test
```

Check TypeScript

```
cd frontend && npm run type-check
```

Manual Testing

1. Create teacher profile via API
 2. Create teaching event
 3. Register student for event
 4. Cancel event and verify refund
 5. Check email notifications sent
 6. Verify payout request flow
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Decisions Made

- Affiliate System: Enable it (uncomment fields in TeachingEvent.ts)
- Dashboard Priority: Events + Bookings first, then Profile, then Payments

Open Questions

1. Teacher Verification: Manual admin approval or automated document verification?
2. Subscription Required: Can teachers create events without subscription?
3. Payout Frequency: Daily/weekly/monthly - configurable per teacher?
4. Multi-currency: Support multiple currencies for teachers in different regions?
5. Co-teachers: Allow multiple teachers per teaching event?